

Do AI centers that use a lot of water need clean water or could it be an added stop on the way to the filtration center?

Posted by u/juulinthepool • 0 points • 9 comments

If it is just used as a cooling element (I think?) would this be possible? Would it help with the strain on water it's causing? Alternatively could they only be built in the ocean so that the ocean water could cool it down or would that heat the water nearby?

Comments

u/Indemnity4 • 1 points • 2025-10-24 06:20 UTC

> could they only be built in the ocean so that the ocean water could cool it

Not a good idea.

You still need to treat the sea water with a bunch of chemicals. You then release all of those back into the ocean. That requires careful planning and consideration.

You can use ocean water for cooling but it's really expensive. Sea water is really corrosive to metal. There are some power generation stations or heavy industry that use ocean water for cooling towers.

u/sexrockandroll • 1 points • 2025-10-24 02:14 UTC

Plumbing for untreated water is a disaster. Plumbing for treated water is a little bit of a disaster already.

u/InsightTussle • 1 points • 2025-10-24 02:08 UTC

They need clean water, but not necessarily drinking quality

u/MrWedge18 • 2 points • 2025-10-24 02:08 UTC

The water has to be extra treated. Usually biocides are added to prevent growth. And the water has to move through very narrow passages (allows for more contact for heat transfer), so any mineral build up can be extremely detrimental. Any dissolved salts can also cause corrosion and leaks.

The up side is it's not gonna be single use. It's in a loop where it absorbs heat from the servers on one side, then gets cooled back down on the other side. The fact it's stuck in a closed loop for awhile is another reason needs to be really clean when it first goes in.

u/juulinthepool • 1 points • 2025-10-24 02:14 UTC

An amazing and informative answer. Thank you. How does the water they use compare to a water park or warehouse or something the same size? Using new water every day vs only having to get a bunch of water all at once? Is it so much that it would have to be multiple years still to even use that amount of water and that's why it is so bad for the water supply or is it just because it is so much all at once and if it is bc it's all at once how is it different?

u/MrWedge18 • 1 points • 2025-10-24 03:53 UTC

[Found a neat infographic by microsoft](https://datacenters.microsoft.com/wp-content/uploads/2023/05/Azure_Modern-Datacenter-Cooling_Infographic.pdf).

A big part I forgot about is the whole "gets cooled back down on the other side" thing. And there are data centers that cool the servers with just regular old fans. So the missing piece is how do you cool the water in the loop, or just the air in the data center.

Apparently the most effective way is evaporative cooling. Any liquid transitioning into gas requires a bit of energy called "latent heat" (this is different from the heat needed to reach the boiling point). Even if it's just evaporation and not boiling, the water still needs to absorb that latent heat. But since the water is evaporating and just being released, it's single use. You need a need to constantly replenish.

Obviously that uses a lot of water. But the upside is it doesn't use a ton of power the way a bunch of A/Cs would.

u/beckdawg19 • 1 points • 2025-10-24 01:53 UTC

It absolutely needs to be clean and non-salinated or it risks ruining the machinery.

u/tmahfan117 • 5 points • 2025-10-24 01:52 UTC

Yes you want treated water in your cooling system. The last thing you want is algae or bacteria or debris clogging up your multi million dollar cooling system that is supporting your hundred million dollar data center

u/juulinthepool • 1 points • 2025-10-24 01:55 UTC

That's fair. It makes sense that they wouldn't choose to use dirty water but if hypothetically regulations were put into place that it had to be even just grey water would that work? Could rain collecting help? Or since the water would heat up I guess it might constantly need new cooled water and couldn't be ran thru the system multiple times.

u/Indemnity4 • 1 points • 2025-10-24 06:12 UTC

Recycled water is the best option. It's usually cheap, about 50-80% the price of regular towns water.

Downside. That does require having a water treatment facility that is making grey water, and have a pipeline going near the facility. Not the easiest thing in the world.

Rain is actually surprisingly dirty. It's open to the air, so it collects a bunch of dust, pollen, and all the various microbes in the air. Microbes + stuff = sludge. So you are back to water treatment again. Much easier to buy pre-treated water from regular facilities.